FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014: 2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

ALL INDIA INTEGRATED TEST SERIES

FIITJEE

JEE(Main), 2015

ANSWERS, HINTS & SOLUTIONS

HALF COURSE TEST–V
(Main)

Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	C	C	C
2.	A	A	B
3.	B	D	D
4.	B	B	A
5.	B	C	D
6.	B	A	A
7.	B	C	B
8.	A	D	D
9.	C	C	B
10.	D	A	B
11.	A	B	D
12.	B	B	A
13.	B	C	A
14.	A	D	A
15.	D	D	A
16.	C	A	B
17.	B	C	C
18.	C	D	A
19.	A	D	A
20.	B	C	B
21.	D	B	C
22.	C	D	D
23.	B	B	B
24.	B	C	C
25.	B	C	D
26.	B	A	C
27.	C	D	A
28.	D	B	A
29.	C	D	A
30.	B	A	C

Physics

PART – I

SECTION – A

1. Kepler's IIIrd law may be written as $T^2 = \frac{4\pi^2}{GM}R^3$

$$v = \frac{2\pi R}{T},\ v^2 \propto \frac{1}{R}$$

2. Buoyant force = Weight of liquid displaced = $\frac{4}{3}\pi r^3 \rho g$

Viscous force = Stoke's drag force = $6\ \pi\eta rv$

$\therefore \quad 6\pi\eta rv = \frac{4}{3}\pi r^3 \rho g$

3. The oscillations in one direction are governed by the equation,
$m\ddot{x} = -kx$
in the other by,
$m\ddot{x} = -4kx\sin^2 60^0$

4. Superposing them gives $2A \sin (kx + \pi/6) \cos (\omega t + \pi/6)$
Nodes are given by $kx + \pi/6 = n\pi$

5. $\alpha = \frac{\tau}{I} = \frac{mgr}{2mr^2} = \frac{g}{2r}$

$a = \alpha r = \frac{g}{2}$.

6. Gravitational potential inside a shell = constant

7. Let A_1 and A_2 be the cross–sectional area of the pipe at points P and Q respectively.
Let v_1 and v_2 be the velocities of oil at the points P and Q respectively.
By conservation of mass,

$$Q = A_1v_1 = A_2v_2$$

$$\Rightarrow \quad v_2 = \left(\frac{A_1}{A_2}\right)(v_1) = 4v_1$$

Applying Bernoullis's equation between points P and Q, we have

$$P_1 + \frac{1}{2}\rho v_1^2 = P_2 + \frac{1}{2}\rho v_2^2$$

$$P_2 = P_1 + \frac{1}{2}\rho(v_1^2 - v_2^2)$$

$$= P_1 + \frac{1}{2} \times 0.9[16 - 256] \times 10^3$$

$$= 280 \times 10^3 + (-108) \times 10^3$$

$P_2 = 172 \times 10^3\ Nm^{-2}$
$P_2 = 172\ (KN\ m^{-2})$.

8. By conservation of momentum
$J = mV + 2\ mV$
$V = J/3m$.

9. $I = \frac{1}{2}mr^2 + \frac{1}{3}m(2r)^2$

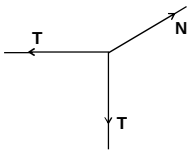
10. Within the liquid, the acceleration is g upward, outside it is g downward.

11. $dQ = dU - dW$, again $dQ = dU + dW$
$dW = pdV = 0$

12. Conservation of total mechanical energy

13. $mg - T = ma$

$T = ma \Rightarrow T = \frac{mg}{2}$ or $N = \frac{mg\sqrt{2}}{2}$



14. At extreme position velocity of M will be zero.

15. At the highest point velocity $v = 0$ after $t = \frac{u}{2g\sin\theta}$ and it will not return to point O.

So distance covered before coming to rest = $\frac{u^2}{4g\sin\theta}$

16. Tension in AB, $2T < 6\,mg$
$\Rightarrow$ Net downward force
$\Rightarrow$ C.O.M. has a downward acceleration.

17. The contact point has a tendency to slide in forward direction, so direction of friction will be in backward direction.

18. $ma + mg - T = ma \Rightarrow T = mg$

19. Normal force by the surface and frictional force an impulsive in nature.
Hence COM is applicable for m only along common tangent.

20. For $y = a\sin\omega t$, $v = a\omega\cos\omega t$
$\therefore$ Acceleration $f = -a\omega^2\sin\omega t$

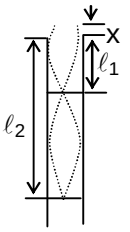
$$\frac{f^2}{\omega^2} + v^2 = a^2\omega^2$$

21. 1 – 2 is isothermal with volume increasing
2 – 3 is isobaric
$\Rightarrow v/T = k$
$\Rightarrow v = kT$
Change in number of moles of the gas does not affect the nature of graph.

22. $\ell_1 + x = \lambda/4$

$\ell_2 + x = \frac{3\lambda}{4}$

$\Rightarrow x = \frac{\ell_2 - 3\ell_1}{2}$



23. $PV = nRT$

$PV = \frac{m}{M}RT$ $\Rightarrow$ slope is directly proportional to the mass of the gas.

30. Use the definition of an elastic collision.

Chemistry

PART – II

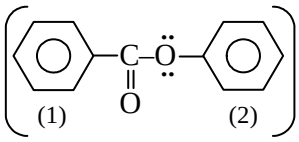
SECTION – A

1. $K_a = Cx^2$
$10^{-4} = 10^{-2} \times x^2$
$x = 0.1$
$i = 1 + x =$ **1.1**

2. $CaCO_3 \xrightarrow{\Delta} CaO + CO_2(X)$
So, X is CO_2
$Na_2CO_3 + H_2O + CO_2 \rightarrow 2NaHCO_3$
Moles of $CaCO_3 = \frac{10}{100} =$ moles of CO_2
Mole of $Na_2CO_3 = 1 mole$
So, moles of $NaHCO_3$ formed according to the moles of CO_2, which is limiting reagent so moles of $NaHCO_3$ formed will be (0.2) mole.

3. Initial temperature = 300 K at
Let new temperature corresponding to t°C is T K.
Let number of moles present at T K = n
Moles of air escaped at T K = n/3
So, number of moles at 300 K = $n + \frac{n}{3} = \frac{4n}{3}$
Volume and pressure are kept constant
$\therefore \quad \frac{n_1}{n_2} = \frac{T_2}{T_1}$
$\frac{4n}{3} \times 300 = nT$
$\therefore$ T = 400 K
Now r.m.s velocity = $\sqrt{\frac{3RT}{M}} = \sqrt{\frac{3 \times 8.314 \times 400}{28.8 \times 10^{-3}}}$ = **588.6 m sec^{-1}**.

6. $K_c = \frac{[Z]}{[X][Y]}$
$[X] = \frac{1}{2}[Y] = \frac{1}{2}[Z] = a$ (let)
$\therefore$ [Z] = 2a, [Y] = 2a, [X] = a
$\therefore \quad 10^4 = \frac{2a}{a \times 2a}$
$a = 10^{-4}$
$\therefore$ [Z] = 2a = **2 × 10^{-4} mol L^{-1}**.

8. In phenyl benzoate  the 1st ring gets deactivated, while 2nd ring gets activated due to lone pairs on oxygen atom. So it can give ortho or para bromo product. But the ortho position is sterically hindered, so it will give para compound as a major product.

9. Staggered conformer of n–butane is achiral because it has a plane of symmetry as well as a centre of symmetry.

10. Basic strength of the oxides of alkali metals increases down the group.

14. $t_{1/2} = \frac{0.0693}{k_1}$ ... (1)

Also, $t_{93.75} = \frac{2.303}{k_1}\log\frac{100}{100-93.75}$

$= \frac{2.303}{k_1}\log\frac{100}{6.25} = \frac{2.303}{k_1}\log 2^4 = \frac{4\times 2.303\times\log 2}{k_1} = \frac{4\times 0.693}{k_1} = 4t_{1/2}$

Hence, (D).

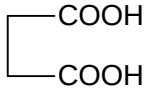
15. Weak base groups are the best leaving groups also stronger is an acid weaker its conjugate base. The CH_3OH is the weakest base among the given group.

16. $e^{-E_a/RT} = \frac{10^{-5}}{100}$

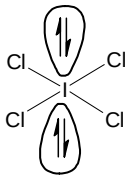
$\therefore \frac{-E_a}{RT} = \ell n\,\frac{10^{-5}}{100}$

$\therefore E_a = 39.94$ kJ

20. In the presence of Zn, only Br_2 can be removed, not as HBr
In the presence of $(CH_3)_2S$



can not be formed (reductive ozonolysis)

21. $\bar{N}=\overset{+}{N}=\bar{N}$ sp ;  sp^3d^2

23. Given $_8B$ and $_7A$ are isobars and thus should have same mass number.
$_8B^m$ and $_7A^m$:
Now, also given $_9C^{17}$ is isotone to $_8B^m$.
$\therefore$ Number of neutrons in both should be same.
or, 17 – 9 = m – 8
m = 16
Mass number of A and B are 16.

24. $6Cl_2 + 10e^- \longrightarrow 2Cl^{5+} + 10Cl^-$

Hence, (C).

26. Let, NH_3 = x mole

N_2H_4 = y mole

$2NH_3 \rightarrow N_2 + 3H_2$

x 0 0

0 $\frac{x}{2}$ $\frac{3x}{2}$

$N_2H_4 \rightarrow N_2 + 2H_2$

y 0 0

0 y 2y

Initial total mole = x + y

Final total mole = 2x + 3y

$\frac{2x+3y}{x+y} = \frac{4.5}{(0.5)\times(4)}$

$\Rightarrow x = 3y \Rightarrow \%x = 75$

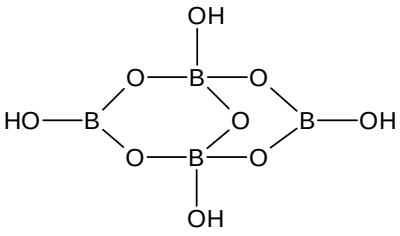
27. $\Delta S = C_v \ln\left(\frac{T_2}{T_1}\right) + R\ln\frac{V_2}{V_1}$

$= C_v \ln 2 + R \times \ln 1/2$

$= (C_v - R)\ln 2$

28. (B) don't have any symmetry.

30.



Mathematics

PART – III

SECTION – A

1. Angle between the asymptotes of $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ is $2\tan^{-1}\left(\frac{b}{a}\right)$ (formula)

Given θ is angle between asymptotes of hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$, and e be its eccentricity

According to question $\theta = 2\tan^{-1}\left(\frac{b}{a}\right) \Rightarrow \frac{\theta}{2}=\tan^{-1}\frac{b}{a} \Rightarrow \tan\frac{\theta}{2}=\frac{b}{a}$

$$\cos\theta=\frac{1-\tan^2\frac{\theta}{2}}{1+\tan^2\frac{\theta}{2}}=\frac{1-\frac{b^2}{a^2}}{1+\frac{b^2}{a^2}}=\frac{a^2-b^2}{a^2+b^2} \Rightarrow 2\cos^2\frac{\theta}{2}-1=\frac{a^2-b^2}{a^2+b^2}$$

$$2\cos^2\frac{\theta}{2}=\frac{a^2-b^2}{a^2+b^2}+1=\frac{a^2-b^2+a^2+b^2}{a^2+b^2}=\frac{2a^2}{a^2+b^2}$$

$$\cos\frac{\theta}{2}=\frac{1}{\sqrt{1+\frac{b^2}{a^2}}}=\frac{1}{\sqrt{e^2}}=\frac{1}{e}$$

2. Any point on the line x + y = 4 can be taken as $(x_1, 4 - x_1)$ as it is a limit distance from the line $4x + 3y - 10 = 0$ we get $\frac{|4x_1+3(4-x_1)-10|}{\sqrt{25}}=1 \Rightarrow |x_1 + 2| = 5$

$\Rightarrow x_1 + 2 = \pm 5$, $x_1 = 3$ or -7

$\therefore$ The required sum = $3 + (-7) = -4$

3. Let $A = x^2 + x + 1$

$\Rightarrow A \in \left[\frac{3}{4}, \infty\right)$

$\Rightarrow (A + 1)^2 - (a - 3)A(A + 1) + (a - 4)A^2 = 0$

$\Rightarrow A^2 + 2A + 1 - (a - 3)(A^2 + A) + (a - 4)A^2 = 0$

$\Rightarrow A(2 - a + 3) + 1 = 0$

$\Rightarrow A(5 - a) = -1$

$$\Rightarrow A=\frac{1}{a-5}$$

$$\Rightarrow \frac{1}{a-5}\geq\frac{3}{4}$$

$$\Rightarrow \frac{19-3a}{a-5}\geq 0$$

$$\Rightarrow a\in\left(5, \frac{19}{3}\right]$$

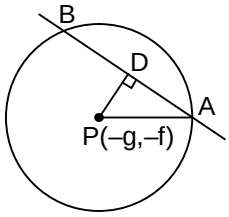
4. $z = x + iy$ then line becomes $(x + iy)\bar{a} + (x - iy)a + b = 0$

$\Rightarrow x(a + \bar{a}) + y(\bar{a} - a)i + b = 0$

$\therefore$ Slope = $\frac{(a+\bar{a})}{i(a-\bar{a})}=-\frac{2Re(a)}{i-2i\{Im(a)\}}=-\frac{Re(a)}{Im(a)}$

5. Total function = 4^5
Total functions when any one element is the left out = ${}^4C_1\, 3^5$
Total functions when any two elements are left out = ${}^4C_2\, 2^5$
Total functions when any three of elements are left out = ${}^4C_3\, 1^5$
Thus the total number of onto functions = $4^5 - ({}^4C_1\, 3^5 + {}^4C_2\, 2^5 - {}^4C_3) = 240$

6. Let D(x, y) be the mid–point of chord AB, we have



$\angle BPA = \frac{2\pi}{3}$, $\angle DPA = \frac{\pi}{3}$

$\Rightarrow PD = PA\cos\frac{\pi}{3}$

$\Rightarrow 4PD^2 = PA^2$

$\Rightarrow 4[(x + g)^2 + (y + f)^2] = g^2 + f^2 - c$

$\Rightarrow 4x^2 + 4y^2 + 8gx + 8fy + 3g^2 + 3f^2 + c = 0$

7. Equation of normals at any point $P(x_1, y_1)$ on the $\frac{a^2(x - x_1)}{x_1} = \frac{b^2(y - y_1)}{y_1}$

If it passes through Q(h, k) then $a^2y_1(h - x_1) = b^2x_1(k - y_1)$ thus foot of normals drawn from Q(h, k) to the ellipse will lie on the hyperbola, whose equation will be

$b^2xk - a^2hy + (a^2 - b^2)xy = 0$ ..... (1)

In the given question if the normals are concurrent at Q(h, k) then

$\frac{x^2}{a^2} + \frac{y^2}{b^2} - 1 + \lambda(\ell x + my - 1)(px + qy - 1) = 0$ ..... (2)

Equation (2) will represent hyperbola given by (1) by comparing the coefficients $a^2\ell p = b^2mq = 1$

8. $f(x) > 0, \forall x \in R$

$\therefore$ Required area $A = \int_0^1 (ax^2 + bx + c)\,dx$

$A = \frac{a}{3} + \frac{b}{2} + c = \frac{1}{6}[2a + 3b + 6c]$

Now, f(0) = c, f(1) = a + b + c

f(–1) = a – b + c

$a = \frac{1}{2}\{f(1) + f(-1)\} - f(0)$ and $b = \frac{1}{2}\{f(1) - f(-1)\}$

$\therefore \Delta = \frac{1}{12}\{5f(1) - f(-1) + 8f(0)\}$

9. $\sin^3x \sin 3x = \frac{1}{2}\sin^2 x(\cos 2x - \cos 4x)$

$= \frac{1}{2} \times \frac{1}{2}(1 - \cos 2x)(\cos 2x - \cos 4x)$

$= \frac{1}{4}(\cos 2x - \cos 4x - \cos^2 2x + \cos 2x \cos 4x)$

$= \frac{1}{4}\left[\cos 2x - \cos 4x - \frac{1 + \cos 4x}{2} + \frac{1}{2}(\cos 6x + \cos 2x)\right]$

$= \frac{1}{4}\left[\frac{3}{2}\cos 2x - \frac{3}{2}\cos 4x + \frac{1}{2}\cos 6x - \frac{1}{2}\right]$

$\Rightarrow n = 6$

10. $I = \sum_{r=1}^{18} \sin^2 (5r)^o = \sum_{r=0}^{18} \sin^2 (5r)^o = \sum_{r=0}^{18} \sin^2 \{5(18-r)\}^o$

$I = \sum_{r=0}^{18} \sin^2 (90-5r)^o = \sum_{r=0}^{18} \cos^2 (5r)^o$

$2I = \sum_{r=0}^{18} (\sin^2 5r + \cos^2 5r) = 19$

$\Rightarrow I = \frac{19}{2}$

11. $\tan^{-1}\left(\frac{\tan x}{4}\right) + \tan^{-1}\left(\frac{3\sin 2x}{5+3\cos 2x}\right)$

$= \tan^{-1}\left(\frac{\tan x}{4}\right) + \tan^{-1}\left[\frac{6\tan x}{8+2\tan^2 x}\right] = \tan^{-1}\left(\frac{\tan x}{4}\right) + \tan^{-1}\left[\frac{3\tan x}{4+\tan^2 x}\right]$

$= \tan^{-1}\left[\frac{16\tan x + \tan^3 x}{16+\tan^2 x}\right] = \tan^{-1}(\tan x) = x$

12. $a + b + c + d = 2$

$AM \geq GM$

$\frac{(a+b)+(c+d)}{2} \geq [(a+b)(c+d)]^{1/2}$

$1 \geq [(a+b)(c+d)]^{1/2} \Rightarrow (a+b)(c+d) \leq M$

$\therefore (a+b)(c+d) > 0$

So $0 < M \leq 1$

13. If $1^2 + 2^2 + 3^2 + ... + 2003^2 = (2003)(4007)(334)$

$(1)(2003) + (2)(2002) + 3(2001) + ... + (2003)(1) = (2003)(334)(x)$

$\Rightarrow \sum_{r=1}^{2003} r(2003-r+1) = (2003)(334)(x)$

$\Rightarrow 2004(\sum r) - (\sum r^2) = (2003)(334)(x)$

$\Rightarrow 2004\frac{(2003\times 2004)}{2} - (2003)(4007)(334)$

$\Rightarrow (2003)(334)x \Rightarrow x = 2005$

14. $\sum_{r=0}^{n} {}^nC_r a^r b^{n-r} \cos\{rB-(n-r)A\} = \text{Re}\left[\sum_{r=0}^{n}\left\{{}^nC_r a^r b^{n-r} e^{i\{rB-(n-r)A\}}\right\}\right]$

$= \text{Re}\left[\sum_{r=0}^{n} {}^nC_r \left(ae^{iB}\right)^r \left(be^{-iA}\right)^{n-r}\right] = \text{Re}\left[ae^{iB} + be^{-iA}\right]^n$

$= \text{Re}(a\cos B + ia\sin B + b\cos A - ib\sin A)^n$

$= \text{Re}(a\cos B + b\cos A)^n = c^n$

15. It is same as dividing 22 books in 5 packets such that two packets have 5 books, each and remaining three have four books each and then the distributing these packets thus total ways fo distributing these books $= \frac{22!}{(5!)^2(4!)^3 2!3!}(5!) = \frac{22!}{2!3!5!(4!)^3}$

16. $\left(\frac{z-1}{z+1}\right)=\left(\frac{x+iy-1}{x+iy+1}\right)=\left\{\frac{(x-1)+iy}{(x+1)+iy}\right\}\times\left\{\frac{(x+1)-iy}{(x+1)-iy}\right\}=\frac{x^2+y^2-1}{(x+1)^2+y^2}+\frac{2iy}{(x+1)^2+y^2}$

Now, $\text{agr}\left(\frac{z-1}{z+1}\right)=\frac{\pi}{4}$

$\therefore\ \tan^{-1}\left[\frac{2y}{(x+1)^2+y^2}\times\frac{(x+1)^2+y^2}{\left(x^2+y^2-1\right)}\right]=\frac{\pi}{4}$

$\Rightarrow x^2 + y^2 - 2y = 1$

17. $(1 + \omega)(1 + \omega^2)(1 + \omega^4)(1 + \omega^8)$ ..... 2n term $\quad(\because\ \omega^3 = \omega^6 = 1)$

$= (1 + \omega)(1 + \omega^2)(1 + \omega^3\omega)(1 + \omega^6\omega^2)$ ..... = [(1 + ω)(1 + ω) ..... n term][1 + ω²)(1 + ω²) ..... n term]

$= (1 + \omega)^n(1 + \omega^2)^n = [(1 + \omega)(1 + \omega^2)]^n = 1^n = 1$

18. $i^i = e^{i\log i} = e^{i[2n\pi i+\log i]} = e^{i\left[2n\pi i+\frac{1}{2}\log i+i\tan^{-1}\left(\frac{1}{0}\right)\right]}$

$= e^{i\left[2n\pi i+i\frac{\pi}{2}\right]} = e^{-(4n+1)\frac{\pi}{2}}$ (where n = 0, 1, 2, 3 .....)

Put n = 0, 1, 2, 3, ... then $i^i = e^{-\frac{\pi}{2}}$, $e^{-\frac{5\pi}{2}}$, $e^{-\frac{9\pi}{2}}$ ..... are in G.P.

$\therefore$ Common ratio = $e^{-2\pi}$

19. $\cos\left[\tan^{-1}\left\{\sin\left(\cot^{-1}x\right)\right\}\right]$

$=\cos\left[\tan^{-1}\left\{\sin\left(\sin^{-1}\frac{1}{\sqrt{1+x^2}}\right)\right\}\right]=\cos\left[\tan^{-1}\frac{1}{\sqrt{1+x^2}}\right]$

$=\cos\cos^{-1}\left(\frac{\sqrt{1+x^2}}{\sqrt{2+x^2}}\right)=\sqrt{\frac{x^2+1}{x^2+2}}$

20. $f[g(x)] = f\left[\frac{3x+x^3}{1+3x^2}\right] = \log\left(\frac{1+x}{1-x}\right)^3 = 3\log\left(\frac{1+x}{1-x}\right) = 3f(x)$

21. $\log (1 + x + x^2 + ...) = \log (1 - x)^{-1} = -\log (1 - x)$

$\Rightarrow A = -1$

$\Rightarrow \cos\theta = -1$

$\Rightarrow \theta = \pi$

22. No value of x

23. $z^7 + 1 = 0$

$\Rightarrow z^7 = -e^{i(\pi + 2k\pi)}$ (where k = 0, 1, 2, ... 6)

$z = -1,\ e^{\pm i\frac{\pi}{7}},\ e^{\pm i\frac{3\pi}{7}},\ e^{\pm i\frac{5\pi}{7}}$

$z^7+1=[z-(-1)]\left[z-e^{i\frac{\pi}{7}}\right]\left[z-e^{-i\frac{\pi}{7}}\right]\left[z-e^{i\frac{3\pi}{7}}\right]\left[z-e^{-i\frac{3\pi}{7}}\right]\left[z-e^{i\frac{5\pi}{7}}\right]\left[z-e^{-i\frac{5\pi}{7}}\right]$

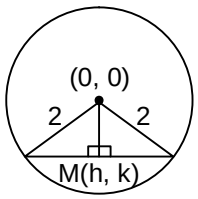
Put $z^2 + 1 = 0 \Rightarrow z = \pm i$

$-8\cos\frac{\pi}{7}\cos\frac{3\pi}{7}\cos\frac{5\pi}{7}=1$

$\Rightarrow\ \cos\frac{\pi}{7}\cos\frac{3\pi}{7}\cos\frac{5\pi}{7}=-\frac{1}{8}$

26. $x^2 + y^2 + 2gn + 2fg + c = 0$ is the equation of circle
$\because$ (a, 0), (0, b) and (a, b) passes through the circle
$\therefore a^2 + 2ag + c = 0$
$b^2 + 2bf + c = 0$
$a^2 + b^2 + 2ag + 2bf + c = 0$
$g = -\frac{a}{2}$, $f = -\frac{b}{2}$, $c = 0$
Radius = $\frac{1}{2}\sqrt{a^2 + b^2}$

27. Equation of chord $T = S_1$
$\Rightarrow xh + ky = h^2 + k^2$ ..... (1)
$\Rightarrow xh + hk - (h^2 + k^2) = 0$
$x^2 - 2x - 2y = 0$
$\Rightarrow x^2 - 2x - 1 - 2y - 1 = 0$
$\Rightarrow x^2 - 2x\left[\frac{hx + ky}{h^2 + k^2}\right] - 2y\left[\frac{hx + ky}{h^2 + k^2}\right] = 0$
$\Rightarrow h^2 + k^2 - 2h - 2k = 0$
$\Rightarrow x^2 + y^2 - 2x - 2y = 0$



28. $(1 + x)^n = C_0 + C_1x + C_2x^2 + C_3x^3$ .....
$C_0 + 3C_1 + 5C_2 +$ ..... $(2n + 1)C_n = (C_0 + C_1 + C_2 +$ ..... $+ C_n) + 2(C_1 + 2C_2 + 3C_3 +$ ..... $+ {}^nC_n)$
$= 2^n + n\,2^n \Rightarrow (n + 1)\,2^n$

29. $D > 0$
$\Rightarrow (a + b + c)^2 - 3(ab + bc + ca)\lambda > 0$
$\Rightarrow 3\lambda - 4 < 0$, $\left[\frac{a^2 + b^2 + c^2}{ab + bc + ca} < 2\right]$
$\Rightarrow \lambda < \frac{4}{3}$

30. $S_n = (1 + 2T_n)(1 - T_n)$
Put n = 1 then $T_1 = (1 + 2T_1)(1 - T_1)$
$\Rightarrow T_1 = \frac{1}{\sqrt{2}}$
Put n = 2 we get $T_2^2 = \frac{1}{2} - \frac{1}{2\sqrt{2}} = \frac{\sqrt{4} - \sqrt{2}}{4} = \frac{\sqrt{a} - \sqrt{b}}{4}$
$\Rightarrow (a + b) = 6$